

SPPA: A Deterministic Runtime Contract for Semantic Primitive Proxy Actors in UAV Digital Twins

Authors omitted for review

Abstract

Object-level UAV telemetry is not directly renderable: a detector may provide a class label, confidence, track identifier, bounding region, and approximate world pose, but no mesh. This paper introduces Semantic Primitive Proxy Assembly (SPPA), a deterministic semantic actor contract that converts sparse object telemetry into low-polygon, role-labeled primitive parts when curated 3D assets are unavailable, too expensive, or semantically mismatched.

SPPA separates runtime representation from visual asset generation. Reviewed archetype recipes map detector labels to part descriptors such as cab, cargo, wheels, trunk, canopy, torso, or fallback volume; runtime packets then update pose, confidence, scale evidence, and material tags without regenerating mesh topology when the part layout is unchanged. The current implementation includes a finite-archetype prototype generator, explicit unknown-label fallback, a SPPA-DESC-0.2/SPPA-UPD-0.2 descriptor/update contract, and an optional Unreal Engine 5.7 backend switchable against curated assets using the same obstacle input. SPPA does not promise arbitrary perfect reconstruction; it converts imperfect semantic evidence into lightweight, controllable, safety-oriented 3D proxies. Local measurements show millisecond-scale descriptor creation in the typical case, cheap update packets, limited packaged desktop feasibility, and clear limits where curated assets or neural generators are better choices. The paper does not claim dense VR readiness, operator benefit, live-flight validation, or a universal bandwidth advantage.

1 Introduction

Semantic-telemetry UAV digital twins replace part of the continuous video stream with object-level messages. A UAV-side perception stack can report object tracks, classes, confidence, positions, and approximate geometry. The ground station then has to represent those objects in an Unreal/Cesium digital twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and runtime loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

SPPA proposes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes. Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The scientific question is therefore not whether SPPA looks better than neural 3D generation. It should not. The question is whether a deterministic, track-persistent, updateable primitive actor is a useful runtime representation for UAV digital twins when the available input is a semantic detection rather than a clean image crop or a curated mesh.

2 Contribution and Scope

The main contribution is a semantic telemetry-to-actor contract, not a new UAV flight pipeline. SPPA consumes the same object input as a conventional Unreal asset-spawning path and produces a switchable alternative representation: curated/static mesh assets or generated semantic primitive actors. This separation keeps the claim narrow: this paper concerns only the object-representation layer that turns semantic detections into runtime actors.

The contribution is divided into four claims:

- **Semantic normalization.** Approximate detector evidence is not copied directly into the scene. It is routed through a SPPA normalizer that maps labels such as `vehicle`, `agricultural vehicle`, or `electric pylon` to reviewed proxy families or conservative fallback.
- **Contract.** A semantic descriptor/update format separates actor creation from pose, material, uncertainty, and matching-topology shape updates.
- **Runtime role.** The same obstacle input can select curated assets or generated primitive actors, so SPPA is an optional representation for uncovered or mismatched labels rather than a replacement for good assets.
- **Evidence boundary.** Current evidence is limited to local resolver, descriptor, synthetic geometry, and Unreal packaged tests; real UAV masks, dense VR, operator benefit, and measured radio-link advantage remain outside the supported claim.

SPPA: language-assisted part decomposition compiled into deterministic 3D primitives

Language can draft the recipe offline; the runtime path uses a reviewed cache entry and telemetry only.

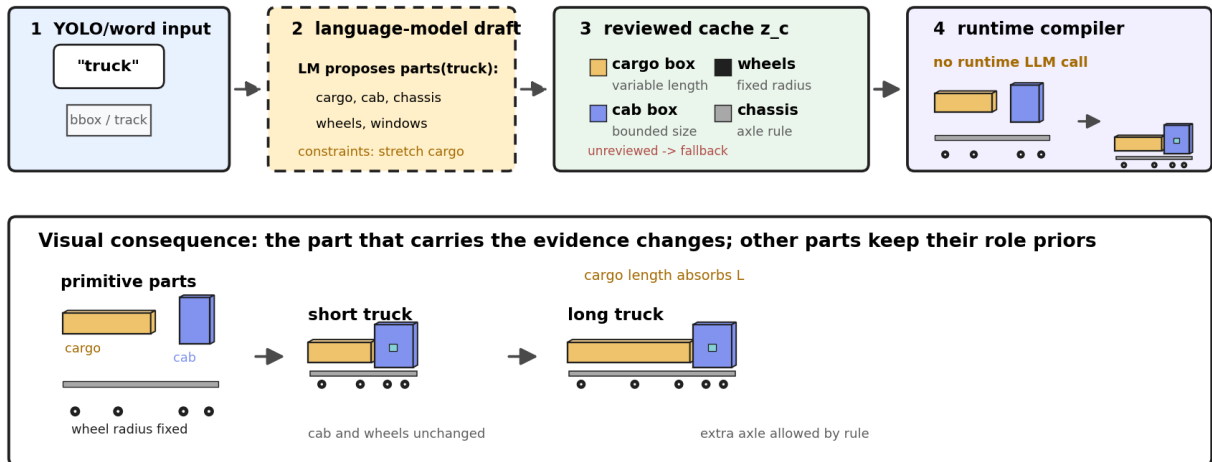


Figure 1: SPPA language-to-parts-to-3D mechanism for the truck archetype. A language model, ontology tool, or human author may draft candidate part roles offline, but runtime execution uses only a reviewed cache entry z_c . The figure shows the proposed chain explicitly: an input label or word activates an offline draft, the draft is reduced to reviewed part roles and bounded update rules, and runtime compiles those roles into primitive geometry. The short and long proxy variants illustrate the implemented role-specific rule: cargo length changes while cab and wheel priors remain bounded, so the object is not represented by root-scaling a single box.

Claim boundary: Figure 1 is not a claim of photorealistic reconstruction or universally correct geometry; SPPA is only claimed here as a bounded, reviewed semantic-proxy compiler.

Figure 1 states the central representation claim visually: SPPA is a reviewed semantic recipe compiler, not arbitrary word-to-3D generation and not a primitive-shape benchmark. The language

component is limited to offline decomposition drafting and ontology expansion. The implemented runtime path starts after that cache exists: a detector label and telemetry state select a reviewed archetype, fill bounded part parameters, and expose explicit fallback behavior when the label is unsupported. For vehicles, the important distinction is part-parametric adaptation rather than global mesh or root-scale deformation.

3 Representation Design Space

The relevant comparison is not a single “3D representation” baseline. A UAV digital twin can visualize a semantic detection through several families of representations, each making different assumptions about input evidence, runtime budget, geometry, and operator interpretation. Geometry-only displays such as boxes, ellipsoids, or billboards remain useful lower-bound controls, but they are not the scientific competitor. SPPA targets the gap between a generic box and a curated asset: a role-labeled primitive actor when only sparse semantic telemetry is available.

Table 1 states the resulting deployment policy. It deliberately gives priority to curated assets and mapping geometry when those are the correct tools.

Table 1: Operational representation policy for semantic detections.

Situation	Preferred representation	Reason	Boundary
Matched project asset exists	Curated asset backend	best appearance and lower measured cost in the cow/tower subset	asset library must match class and scale
Known semantic family, no asset	SPPA primitive actor	persistent, local, role-labeled 3D proxy from label/pose input	finite reviewed archetypes
Unknown or adversarial label	SPPA fallback volume	avoids inventing a specific object from weak semantics	not visually specific
Clean crop plus offline budget	Image-to-3D or asset authoring	richer mesh can be generated outside the flight update loop	no per-frame track contract
Planning or occupancy task	Occupancy / ESDF / map layer	represents navigable space and collision evidence	not a role-labeled display actor

4 Related Work

SPPA overlaps with several mature research areas, but it does not replace them. Text-to-3D and image-to-3D generation methods such as DreamFusion, Magic3D, Point-E, Shap-E, TripoSR, InstantMesh, Stable Fast 3D, SPAR3D, LGM, CRM, Unique3D, TRELLIS/TRELLIS.2, and Hunyuan3D 2.x target visual asset generation or reconstruction [25, 17, 20, 15, 35, 43, 29, 30, 31, 32, 38, 39, 42, 41, 47, 33]. Their strength is fidelity when suitable visual or prompt evidence is available. SPPA targets a different runtime point: deterministic actor generation from sparse semantic telemetry, low polygon counts, and cheap per-frame updates. Recent benchmark work also highlights why a real image-to-3D ranking needs a shared reference set, metrics, and part/structure evaluation rather than a local figure assembled from heterogeneous inputs [34, 44]. For this reason, the artifact package includes an explicit SOTA-protocol readiness manifest and verifier. In the current artifact state that verifier keeps the comparative figure classified as an input-alignment audit, not as an image-to-3D SOTA leaderboard.

Monocular and open-vocabulary 3D detection estimate 3D boxes or object state from images [3, 45]. Animal pose and animal reconstruction methods estimate detailed skeletons or category-specific 3D shape [46, 16, 13]. These methods can provide stronger evidence to SPPA, but SPPA itself is a rendering/backend layer, not a detector or pose estimator.

Primitive representations have a long history. Marr’s volumetric primitives, recognition-by-components, superquadrics, learned part decomposition, and recent superquadric scene decomposition all show that structured primitive geometry can encode meaningful shape abstractions [18, 2, 28, 24, 36, 10]. Procedural modeling and shape grammars show how compact programs can generate structured objects [19, 14]. SPPA uses these ideas operationally, but the claimed novelty is the bounded, track-updateable semantic telemetry contract inside an Unreal UAV digital twin.

Graphics systems already provide level-of-detail, impostors, billboards, streaming tiles, and virtualized geometry [4, 5, 8, 23]. 3D and dynamic scene graphs maintain semantic world structure [1, 27, 40]. Occupancy grids, OctoMap, ESDF, and TSDF maps support planning and mapping [12, 22]. SPPA should be read as a display actor representation that can consume such upstream evidence, not as a mapping substrate.

Finally, any operator-facing claim must be evaluated with human-factors methods. SAGAT, situation-awareness theory, NASA-TLX, and uncertainty-display literature are relevant to future SPPA validation [6, 7, 11]. The present paper contains no pilot study.

5 SPPA Contract

At time t , SPPA consumes an object state

$$x_t = \{i, \ell, c, b_t, m_t, \xi_t, s_t, \Sigma_t, h_t\},$$

where i is a track identifier, ℓ a detector label, c confidence, b_t a bounding region, m_t an optional mask or footprint, ξ_t a world pose estimate, s_t optional metric dimensions, Σ_t uncertainty, and h_t the track history. The resolver maps ℓ to one of three cases:

- **exact_class**: implemented label with a fixed archetype;
- **keyword_archetype**: out-of-template label mapped to a finite reviewed archetype, such as vehicle or vertical structure;
- **fallback_unknown_label**: unknown label rendered as a generic uncertainty-aware volume.

This is deliberately not universal word-to-3D generation. A language model may assist offline ontology expansion, but runtime resolution must remain bounded, reviewed, cached, and able to reject over-specific mappings. Figure 1 makes this boundary explicit: language-model support belongs to offline archetype authoring or review, while the runtime path maps a detector label to a reviewed archetype, generic fallback volume, descriptor parts, and Unreal components. The current implementation includes a versioned archetype-recipe contract with 15 reviewed archetypes and 64 exact/keyword/fallback label checks. That contract verifies resolver status, allowed runtime archetype, required semantic material roles, and disabled runtime LLM use; it does not validate geometric correctness for arbitrary open-vocabulary labels. A separate offline recipe-review gate now tests five candidate mappings before caching: one accepted reviewed-family mapping, one fallback-only mapping, and three rejected mappings for visual artifact context, unreviewed topology/material roles, or attempted runtime LLM use. This is an implementation guard, not proof of open-vocabulary coverage.

5.1 Part Graph

Each archetype a defines a part graph

$$G_a = (V_a, E_a), \quad v_k = (p_k, r_k, \tau_k, \rho_k),$$

where p_k is a primitive type, r_k is its role, τ_k its transform relative to the actor root, and ρ_k role-specific parameters such as dimension priors, color/material role, and confidence tags. The current implementation uses boxes, cylinders, ellipsoids, cones, and tori.

Vehicle adaptation is the clearest implemented case. A long truck and a short truck should not be produced by globally scaling the same box. In the current vehicle grammar, cab and wheel dimensions remain bounded by role priors, while the cargo/body span and axle placement absorb length changes. In the synthetic truck check, two descriptors with equal width/height and different length keep cab scale fixed at [1.656, 2.070, 1.952] m and wheel scale fixed at [0.392, 0.094] m, while cargo/body length changes from 2.263 m to 5.188 m. This is an internal part-invariance diagnostic, not a visual comparison against competing 3D generation methods and not proof that SPPA recovers true truck morphology from UAV imagery. The corresponding component views are kept in the artifact log rather than used as main-paper comparative evidence.

5.2 Pose, Yaw, and Updates

The descriptor separates actor pose ξ_t from part parameters q_t . Pose changes are per-frame updates; topology regeneration is avoided unless the class/archetype changes or new evidence crosses a configured threshold. A typical update packet therefore changes only root transform, confidence, material/uncertainty tags, or a small set of shape parameters.

Yaw ambiguity is handled explicitly. A top-down footprint or PCA orientation gives an axial orientation modulo π , not necessarily a signed heading modulo 2π . Signed yaw requires velocity, asymmetry, multi-frame evidence, or an external heading source. If no reliable sign exists, the descriptor sets `yaw_ambiguous=true`. The exact thresholds and weights in the fitting objective are exposed as versioned protocol parameters in the current implementation, not calibrated constants.

When a mask polygon is available, the descriptor records both the axis-aligned mask box and an oriented image-space footprint from PCA projections. This is used as footprint evidence and axial yaw evidence. If explicit metric dimensions or an explicit image-to-ground scale are supplied, the current generator can convert the calibrated footprint to metric length/width and pass those dimensions to the part-parametric builder. Without such calibration, the mask remains image-space evidence only and does not change metric geometry.

6 Implementation

The prototype has three layers. The first is a Python OBJ/MTL prototype generator for a finite set of explicit and keyword-resolved archetypes, including vehicles, vulnerable road users, vegetation, vertical structures, generic quadrupeds, and unknown fallback volumes. This path is useful for visual inspection and lightweight local timing only.

The second layer emits versioned descriptors and update packets. A `SPPA-DESC-0.2` record stores resolver source, match type, fallback reason, scale source, axis-aligned and oriented mask footprint evidence, yaw ambiguity, shape confidence, material source, optional observed-color evidence, part list, primitive counts, and scheduler metadata. A `SPPA-UPD-0.2` record separates create/regenerate payloads from pose/no-op/shape updates. The design goal is to generate once per track/archetype, update pose/material/uncertainty every frame, and apply shape-parameter changes in place when the part topology is unchanged. In the current generator, `descriptor_id` is a topology-cache identity: it excludes pose, timestamp, confidence, and observed color, which remain in `input_hash` or update fields. This separation is required for cheap per-frame updates; otherwise a pose update would look like a new mesh descriptor.

The third layer is an optional Unreal Engine 5.7 telemetry backend. The same obstacle payload can drive either curated asset actors or semantic proxy actors. The asset backend remains the default. The SPPA backend can ingest descriptor JSON, construct primitive components from `parts[]`, apply pose/no-op/shape update packets, and reject invalid JSON or mismatched descriptor identifiers without changing the current actor. This implements a runtime switch without forcing a migration away from curated assets.

A resolver coverage regression checks ten common or plausible UAV labels: `person`, `bicycle`, `cyclist`, `motorcycle`, `bus`, `delivery van`, `pickup truck`, `dog`, `radio mast`, and an unsupported

object label. All resolve to the expected finite archetype or to the explicit unknown fallback in the current implementation. This is a resolver regression check, not evidence that the ontology is complete.

7 Preliminary Evaluation

All measurements in this section are preliminary operating-point tests. They are reported because they constrain the claim, not because they complete the validation of SPPA as a UAV/VR system.

7.1 OBJ Construction Sanity Check

The original OBJ construction timing covered the initial seven-class prototype set and is retained only as a legacy construction/export sanity check. In that run, per-object export stayed below 0.1 s including Python startup and file export overhead, with meshes ranging from 488 vertices and 396 faces for `biker` to 856 vertices and 786 faces for `truck`. The current generator has since been extended to a broader finite-archetype resolver. These numbers are therefore implementation evidence for the OBJ construction path only, not current-ontology coverage, Unreal instancing timings, or end-to-end detection-to-render latency.

7.2 Benchmark Alignment and Generator Substitutability Stress Test

Table 2 is a substitutability stress comparison, not a SOTA ranking. A real SOTA ranking for image-to-3D would require the same detection crops or reference images, preferably with target meshes or structured human-preference scores, and should include contemporary high-fidelity systems such as TRELLIS.2 and Hunyuan3D 2.1 alongside fast feed-forward methods such as SF3D, SPAR3D, and TripoSR. SPPA would not be ranked as an image-to-3D asset generator in that protocol, because its intended input is a semantic detection state rather than a visual crop. The local stress test asks a narrower systems question: whether generators that were runnable in the local Windows workstation stack can replace a bounded runtime proxy under a portable-flight profile. Neural generators are expected to produce richer assets when they receive appropriate prompts or images; SPPA’s question is persistence, updateability, and deterministic fallback from sparse telemetry.

All neural runs used the same RTX 5090 workstation with a 6 GB PyTorch CUDA allocator budget. This is a stress profile, not a physical GPU partition. It is motivated by 2026 laptop constraints: NVIDIA lists 8, 12, 16, and 24 GB memory configurations across RTX 50-series laptops [21], Epic recommends at least 8 GB graphics RAM for UE5 development [9], and the Steam May 2026 hardware survey reports large installed shares at 8, 12, and 16 GB [37]. The allocator limit follows PyTorch’s documented memory-fraction mechanism [26].

Table 2: Preliminary July 2026 substitutability stress comparison over six object classes. Times are local wall-clock measurements under constrained input settings, not statistical or end-to-end UAV/VR validation.

Method	Input	Wall	Tris	VRAM
SPPA finite archetype	label/tag	0.0016 s med.	488–1,668	0 MB
Shap-E text $K = 16$	prompt/tag	2.0080 s	17,884–131,668	6,120 MB
Point-E text + SDF32	prompt/tag	6.4266 s	1,026–5,454	3,644 MB
TripoSR warm 128 ³	RGBA crop	0.4604 s	19,284–31,012	2,028 MB
Hunyuan3D-2mini Turbo	RGBA crop	1.5042 s	332,020–1,698,032	6,092 MB

The defensible ranking from these data is therefore not a visual SOTA ranking. It is a task-fit ranking for the runtime contract that SPPA claims to satisfy: low-latency generation, bounded

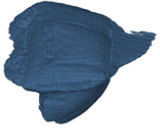
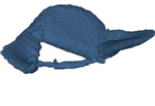
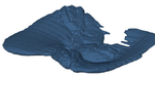
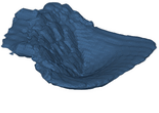
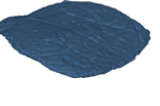
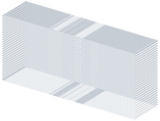
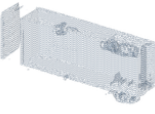
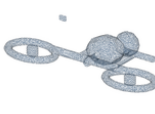
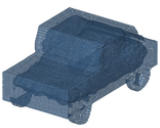
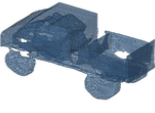
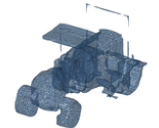
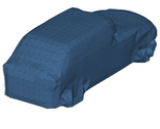
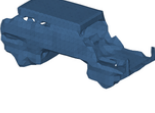
geometry, no neural GPU inference in the runtime loop, deterministic descriptors, and track-updateable actor semantics. Under that contract, Table 3 ranks the reproduced local methods by explicit pass/fail criteria. This ranking should be read together with Figure 2: neural generators can be visually richer when given suitable visual evidence, while SPPA is intentionally optimized for a different runtime representation problem.

Table 3: SPPA runtime task-fit ranking derived from the local July 2026 generator stress runs. This ranking is for the UAV/VR semantic-proxy contract, not for image-to-3D visual SOTA. The six score criteria are: median generation below 33 ms, maximum mesh below 5k triangles, maximum torch reserved memory below 1 GB, no GPU inference, deterministic descriptor contract, and track-updateable actor semantics.

Method	Input	Score	Wall	Max tris	VRAM	Failed criteria
SPPA finite archetype	semantic label/tag	6/6	0.0016s	1,668	0MB	-
TripoSR warm	clean RGBA proxy crop	0/6	0.4604s	31,012	2028MB	lat., tris, mem, GPU, contract, update
Hunyuan3D-2mini Turbo	clean RGBA proxy crop	0/6	1.5042s	1,698,032	6092MB	lat., tris, mem, GPU, contract, update
Shap-E text K=16	prompt/tag	0/6	2.0080s	131,668	6120MB	lat., tris, mem, GPU, contract, update
Point-E text + SDF32	prompt/tag	0/6	6.4266s	5,454	3644MB	lat., tris, mem, GPU, contract, update

Two additional fast image-to-3D candidates are intentionally excluded from the numeric table. Stable Fast 3D installed in a Python 3.10 environment but emitted no benchmark event before the 20 minute timeout in this pass. SPAR3D installation completed, but model loading failed before inference because the required weights were gated. These are unresolved reproducibility outcomes, not measured failures of output quality.

Figure 2 is a qualitative input-alignment audit across the six object cases. The first row shows the local crop/proxy image supplied to image-conditioned methods; it is not real detection ground truth. The final benchmark should replace this row with recorded detection crops, masks, bounding boxes, or a public benchmark reference set, and should report explicit quality metrics. The current image-conditioned methods were given clean proxy RGBA crops, while text-conditioned methods were run in speed-oriented settings. An input-provenance manifest and verifier mark all six current first-row images as synthetic proxy crops with zero ground-truth items. This prevents the figure from being re-labeled as a detector or ground-truth benchmark without new evidence. A separate synthetic YOLO reference manifest currently provides three 2D ground-truth bbox labels (**biker**, **cow**, and **tower**), but those legacy samples are bbox-only for this audit and do not provide readable crops for the six-class visual grid. The point of the figure is narrower: neural generators produce meshes with different fidelity/cost assumptions, while SPPA produces a bounded actor-like primitive proxy from the semantic label.

	car	truck	tractor	biker	cow	tree
Text/tag input	text: "car"	text: "truck"	text: "tractor"	text: "biker"	text: "cow"	text: "tree"
SPPA						
TripoSR						
Hunyuan3D						
Shap-E						
Point-E						

Text/tag row is label-only; RGBA crop inputs are omitted here to avoid proxy-input bias.

Figure 2: Input-aligned visual audit for the six local stress-test cases. The top row is deliberately text/tag only, not a proxy image and not ground truth. SPPA uses the current label-only finite-archetype prototype path; Shap-E and Point-E use text prompts/tags; TripoSR and Hunyuan3D use separate RGBA crop inputs that are omitted from this grid because proxy crops can visually resemble SPPA outputs. This is a qualitative audit artifact, not a SOTA ranking.

Figure 3 adds four real-image stress probes for the cyclist, electrical-tower, tractor, and tractor-with-trailer cases. These user-supplied crops are real input evidence, not 3D ground truth. The test is deliberately dual-input. In the image path, a YOLOE-26s open-vocabulary detector reads the real image and produces approximate semantic evidence. In the tag-only path, SPPA receives a semantic tag/text input without using the image crop. In both cases the semantic text is routed through a SPPA normalizer before proxy selection, so the paper does not depend on YOLOE producing an exact object name. For example, `vehicle` is accepted only as a generic family and rendered with a conservative heavy-vehicle proxy. Shap-E and Point-E receive the same tag/text prompts; TripoSR and Hunyuan3D receive the same low-quality image crops. The result should be read as an input-quality and semantic-normalization boundary check, not as a visual SOTA ranking.

	Input evidence real crop, not GT	Detector probe YOLOE image path	SPPA tag/text path	Shap-E text-to-3D tag	Point-E text-to-3D tag	TripoSR image-to-3D crop	Hunyuan3D image-to-3D crop
biker							
tower							
tractor							
tractor+trailer							

Real-input probes: YOLOE and image-to-3D use images/crops; SPPA, Shap-E, and Point-E use text/tag prompts. Not 3D ground truth.

Figure 3: Real input probes for cyclist, electrical tower, tractor, and tractor-with-trailer cases. The first column is real input evidence, not ground truth. The detector path records YOLOE semantic evidence and reviewed 2D crop provenance; the SPPA path uses normalized semantic tags; Shap-E and Point-E use tag/text prompts; TripoSR and Hunyuan3D use the same low-quality crops. This artifact demonstrates why the protocol must separate detector evidence, normalized tags, crop annotations, text prompts, and ground-truth 3D references.

Table 4: Real-probe semantic normalization. The image-side detector evidence is not treated as exact ground truth; SPPA maps it to a reviewed proxy family or a conservative fallback.

Probe	YOLOE image evidence	SPPA normalized proxy	Tag-only SPPA input
Cyclist road image	person + motorcycle	two_wheeled_rider → biker	cyclist/biker
Mountain tower image	electric pylon	power_tower → vertical_structure	tower/electric pylon
Mountain tractor image	agricultural vehicle	farm_vehicle → farm_vehicle	tractor/farm_vehicle
Tractor with trailer image	vehicle	generic_vehicle → heavy_vehicle	vehicle/trailer

These results narrow the argument. TripoSR and Hunyuan3D are better choices when the goal is high-fidelity asset generation from a usable image crop. Shap-E and Point-E demonstrate that text-conditioned meshes can be produced from the same tag path, but their outputs are not track-persistent proxy contracts. SPPA’s measured advantage is different: no GPU inference, millisecond-scale local proxy creation, bounded primitive geometry, and per-track updateability. Whether that advantage matters to pilots is still unproven.

7.3 Role-Preservation Diagnostic

The main competitor set for SPPA is the set of operational representation alternatives in Table 1, especially curated assets and fast text/image-to-3D generators. Geometry-only displays are

retained only as negative controls and lower-bound runtime references; construction timings for isolated boxes, capsules, or ellipsoids are therefore not reported as a result.

The implemented diagnostic asks whether SPPA preserves semantic part roles while adapting dimensions. For four synthetic vehicle variants, median dimension error was 0.3926 for fixed templates, 0.0000 for trivial global scaling, and 0.0223 for SPPA parametric vehicles. The global-scaling result is deliberately not a win condition: it fits the outer box by deforming all parts together, including cab and wheels. The SPPA diagnostic is therefore not a morphology benchmark. Its supported claim is narrower: role-assigned parts such as cargo, cab, windows, and wheels can follow different update rules instead of collapsing into a scaled copy of the same object. A dedicated short/long truck invariance check makes this concrete: increasing vehicle length from 5.2 m to 8.2 m kept cab scale delta at 0.0, tire scale delta at 0.0, increased cargo length by 2.925 m, and increased tire count from 6 to 8.

A calibrated-mask variant now exercises the same path from observed footprint evidence rather than manually supplied dimensions. Over 64 synthetic rectangle masks with explicit ground scale, the calibrated mask-to-shape benchmark derived metric length/width with 0.000/0.000 m P50/P95 summed dimensional error and kept the uncalibrated control path at `template_prior`. Local geometry build time was 442/480 μ s at P50/P95, and descriptor build time was 297/321 μ s at P50/P95. This verifies only the deterministic contract path *mask polygon + supplied ground scale* $\rightarrow$ *parametric parts*; it is not evidence of real UAV segmentation quality, camera calibration accuracy, or silhouette-to-part assignment under occlusion.

7.4 Descriptor and Unreal Runtime Tests

The descriptor/update benchmark generated descriptors for exact labels, keyword-resolved labels, and unsupported labels. Exact and keyword labels resolved to known finite archetypes; unsupported labels produced unknown fallback descriptors. In the replay benchmark, descriptor build with parts over 81,568 threshold-expanded rows had P50/P95 of 391.8/515.0 μ s. The P99 was 671.2 μ s, but the maximum descriptor-build observation was 360.050 ms. Scheduler decisions had P50/P95/P99 of 9.2/14.2/19.2 μ s, and update-packet builds had P50/P95/P99 of 16.8/75.3/91.0 μ s. Contract validation passed 453 descriptor and update-packet checks. These are Python contract measurements, not Unreal frame time; they support typical-case local contract cost, not a hard real-time guarantee.

A preliminary link-budget model over the same replay separates byte accounting from radio-link validation. If no-op packets are suppressed, measured SPPA JSON update traffic was 25.8–37.4 kB/s across scheduler thresholds; emitting every packet was 78.5–94.0 kB/s. A favorable modeled binary box stream was about 3.27 kB/s, while modeled compressed video at 2/5/10 Mbps corresponds to 250/625/1250 kB/s. This does not show that SPPA is the lowest-bandwidth semantic representation. It shows that SPPA is a richer actor representation with byte cost between sparse boxes and continuous video/mesh transfer under this replay model.

An additional adversarial resolver audit covered 12 compound or misleading labels. Visual-artifact contexts such as `toy truck`, `vehicle shadow`, `dog statue`, and `person poster` were forced to the unknown fallback; equipment-only labels such as `worker helmet` also fell back instead of spawning a person. Compound family labels such as `animal feed trailer`, `horse trailer`, `emergency vehicle`, `road sign`, and `field worker` resolved to reviewed family archetypes. This is a resolver-contract check, not a human-factors validation of fallback interpretation.

The offline recipe-review gate complements this resolver audit. It evaluates candidate language/ontology recipes against the finite primitive vocabulary, known material roles, runtime resolver result, triangle budget, and the rule that no language model may be called in runtime. In the current artifact, the gate accepted a `dump truck` candidate as a reviewed heavy-vehicle mapping, allowed an `unlisted obstacle` only as fallback, and rejected `person poster`, `delivery drone`, and `construction crane` candidates for false semantics, unreviewed topology/material roles, or runtime-LLM dependence. This supports the narrower claim that SPPA has a cache-admission guard for candidate recipes.

A mask-footprint benchmark then evaluated descriptor-level shape evidence over 162 synthetic rectangle masks: three lengths, three aspect ratios, six axial rotations, and three corner-jitter levels. Across clean non-axis-aligned masks, the AABB dimensional error was 50.568/109.808 px at P50/P95, while the PCA oriented footprint recovered 0.000/0.000 px dimensional error and 0.000/0.000 deg axial-yaw error. With jittered non-axis-aligned masks, AABB error was 51.957/114.235 px and oriented-footprint error was 6.615/12.389 px, with 0.158/0.666 deg yaw error. Axis-aligned masks can tie the AABB baseline, and jitter can make AABB slightly better for those aligned cases; the supported claim is therefore limited to extracting oriented image-space footprint evidence from supplied polygon masks, not real UAV silhouette recovery.

Unreal tests then exercised the native backend in three steps. First, an actor-level regression test loaded a real `SPPA-DESC-0.2` fixture with 12 parts, generated 12 static-mesh components, propagated material/evidence/uncertainty tags, accepted valid pose updates, rejected mismatched descriptor updates, rejected invalid JSON without replacing the current proxy, and applied a matching-topology `shape_param_update` while preserving the same component names. Second, component replay used the same `obstacles` array payload shape as `/api/ui/data`. Third, a local-loopback HTTP replay exercised Unreal's HTTP polling branch with the same payload. A recorded-payload adapter then converted seven logged `obstacle_ingest` observations into common asset-compatible, SPPA emit-all, and SPPA changed-only payload streams. Unreal Editor-Cmd consumed those recorded streams through the same component entry point with asset, actor-proxy, and instanced-proxy backends. A later packaged executable run consumed the same streams with rendered frames. This is stronger than a payload-only adapter check, but it remains an internal short-stream replay; it is not live network telemetry, VR, phase-aligned GPU-profiler evidence, or operator validation.

The packaged evidence consists of synthetic density replay, HISM batching diagnostics, and a short recorded-payload replay. The baseline packaged replay, `20260702T122743Z`, used three repetitions, 30 warmup frames, and 120 measured frames per phase. At 100 synthetic obstacles, `semantic_proxy` used 432 components, about 208,864 estimated triangles, and 432 estimated draw calls; pose and shape frame P95 were 17.814 ms and 19.256 ms. This passed a 33.3 ms P95 desktop frame budget in that run but missed a 16.7 ms P95 budget, and phase-aligned GPU-profiler counters were not captured. The allowed claim is therefore limited to packaged desktop feasibility in specific measured cases, not VR readiness.

The recorded-payload packaged run, `20260703T024800Z`, replayed three externally generated JSONL streams through the same asset/proxy backend switch: common asset-compatible payloads, SPPA emit-all payloads, and SPPA changed-only payloads. All 27 applications were accepted. The HISM route had median active draw-group count 6 and frame P95 4.809 ms in this short replay, while the actor proxy had median component count 31 and frame P95 6.531 ms. This is useful integration evidence, but it is still a short internal replay without live network timing, phase-aligned GPU-profiler counters, headset execution, or operator evidence.

The HISM measurements are a separate opt-in diagnostic family, not a replacement for the baseline replay. In `20260702T215340Z`, a one-repetition run with shorter measurement windows, the HISM route reduced the 100-object component and estimated draw-call count from 432 to 11, and reduced `shape_param_update` action P95 from 20.736 ms to 13.270 ms. The 100-object shape-stream frame P95 improved from 42.903 ms to 33.626 ms, which was still above the 30 FPS P95 budget. The HISM update path was then changed to defer render-state dirty marking during batch instance updates and flush once per touched group. A longer HISM-only packaged run, `20260703T031246Z`, used three repetitions, 30 warmup frames, and 120 measured frames per phase at 100 objects. It reported 11 active draw groups, pose-stream frame P95 13.884 ms, shape-stream frame P95 29.715 ms, and shape-stream frame P99 35.021 ms, with 6 of 360 shape frames above 33 ms. This was promising but not robust. A later dense sweep without Unreal CSV profiling, `20260703T033655Z`, tested 100, 250, and 500 objects with the same HISM backend. The 100-object shape-stream frame P95 was 38.657 ms, and an isolated 100-object no-CSV rerun, `20260703T033840Z`, reported 37.126 ms. At 250 and 500 objects, shape-stream P95 rose to 94.018

Table 5: Selected Unreal evidence that constrains the SPPA claim. Values are preliminary and do not include VR headset, live-network, broad curated-asset validation, or phase-aligned GPU-profiler validation. Auxiliary regression checks are kept in the artifact package, not in this summary table.

Test	Result	Main boundary	Claim impact
Backend switch smoke	Same obstacle input selects assets, actor proxies, or HISM proxies; default remains assets	Editor smoke, not operator workflow	SPPA is optional, not a forced replacement
Editor-Cmd recorded replay	27/27 payload applications accepted across asset, actor-proxy, and HISM backends; SPPA emit-all P95 apply: 1.614/3.805/4.049 ms	component ingestion under Editor-Cmd/Null-RHI	recorded streams reach all backend routes
Packaged recorded replay 20260703T024800Z	27/27 payload applications accepted; frame P95 assets/proxy/HISM: 16.551/6.531/4.809 ms; apply P95: 2.566/5.670/3.246 ms	one short replay, no live network, no phase-aligned GPU profiler, no headset	recorded streams run in packaged executable
Packaged synthetic 100-object replay	actor-proxy pose/shape frame P95: 17.814/19.256 ms	synthetic desktop obstacles; misses 60 FPS P95	desktop feasibility only, not VR readiness
Packaged HISM scaling 20260703T033655Z/033840Z	11 active draw groups; 100-object no-CSV rerun shape frame P95: 37.126 ms; 250/500 shape P95: 94.018/186.113 ms	synthetic desktop obstacles; shape updates miss 30 FPS P95 at 100+ objects	batching helps draw pressure, but dense shape updates remain limiting
Packaged HISM partial updates 20260703T040602Z	10% pose and 10% shape changed-track schedule; 500-object pose/shape frame P95: 16.528/29.787 ms with 11 draw groups	synthetic scheduler rate, not flight-derived; dense all-object updates still fail	partial-update contract supports changed-only scheduling, not dense all-track updates
Packaged cow/tower asset subset	project assets vs. SPPA shape frame P95: 6.789 vs. 22.445 ms	narrow two-label subset	curated assets remain better when available

ms and 186.113 ms, respectively. The supported conclusion is therefore narrower: HISM batching keeps the proxy scene at 11 active draw groups across these synthetic densities and create-steady frames remain cheap, but stable 30 FPS P95 shape-update performance is not yet demonstrated even at 100 objects. This still does not support a 60 FPS, headset, live-network, or phase-aligned GPU-profiler claim.

The HISM path was then extended with an explicit **partial_obstacle_update** contract: unmentioned entities are retained, while mentioned entities replace their old instance keys before new parts are upserted. The backend verifier creates two entities, applies a partial shape-update to only one, and verifies 6/6 expected live/HISM instances after the untouched entity is preserved. A new packaged run, 20260703T040154Z, confirmed that dense 100-object shape updates still fail (shape-stream P95 46.151 ms). With shape updates limited to 10% of objects but dense pose updates retained, shape-stream P95 stayed below 33.3 ms through 500 objects, but pose-stream P95 reached 52.474/104.502 ms at 250/500 objects. Under a changed-track schedule with both pose and shape updates limited to 10%, 20260703T040602Z reported 500-object pose/shape frame P95 of 16.528/29.787 ms and no measured hitches above 33 ms. The allowed claim is therefore conditional: partial updates support a changed-only scheduler stress case, but the 10% rate is not yet derived from flight logs and does not prove dense all-object update performance.

Current HISM regressions expose `SEMANTIC_PROXY_INSTANCED` as a live opt-in backend, preserve local pose through the component route, reject mismatched descriptor/update packets, support group-level material and uncertainty tags, provide an opt-in collision policy, and support eight per-instance custom-data floats for resolved color, confidence, evidence source, and uncertainty. In an observed-color stress run at 100 objects, per-instance custom data reduced active HISM components/draw calls from 188 to 11 and reduced create/pose/shape action P95 from 27.792/9.669/18.303 ms to 18.608/7.050/15.351 ms, but did not improve the corresponding shape-stream frame P95 (37.725 vs. 38.136 ms). This supports HISM as an engineering direction for batching, not as completed dense-VR validation; phase-aligned GPU-profiler measurements and operational collision semantics remain open. The same backend verifier now includes a local georeferenced-pose regression: direct HISM payloads using `world_m` and `world_cm`, and component-routed payloads using both `SemanticProxy` and `SemanticProxyInstanced`, place a zero-centered descriptor at [1250,-400,120] cm with yaw 37 deg. This is only local-frame pose preservation, not evidence of live UAV georeferencing, camera calibration, Cesium accuracy, or VR runtime behavior.

A second packaged run adds a narrower real-asset baseline for the two labels with reviewed project static meshes in the repository, `cow` and `tower`. At 50 objects, `project_assets` used 50 static-mesh components and about 167,300 estimated triangles; `semantic_proxy` used 200 components and about 113,200 estimated triangles. The real-asset subset had lower P95 frame times for create, pose, and shape phases, with shape-stream P95 6.789 ms versus 22.445 ms for SPPA. This is not a failure of the SPPA claim: it shows that curated assets are preferable when an appropriate asset exists and loads cheaply. SPPA’s intended gap is uncovered or mismatched semantic telemetry, not replacement of available high-quality assets.

8 Discussion

A strict interpretation is that SPPA currently has one defensible advantage and several unresolved weaknesses.

The advantage is runtime structure under sparse input. If the input is only a YOLO tag, track id, confidence, pose estimate, and rough dimensions, neural image-to-3D cannot use information that does not exist. A text-to-3D generator can hallucinate an asset, but it does not naturally produce a bounded, track-persistent, low-component Unreal actor with explicit yaw ambiguity, fallback reason, scale source, and update policy. SPPA does.

The weaknesses are also clear. Geometry is still based on finite archetypes. Open-label handling accepts any label but does not create correct geometry for any word. Mask/PCA footprint support reaches parametric geometry only when an explicit image-to-ground scale is supplied; otherwise it remains image-space descriptor evidence, and it has not been tested on real UAV imagery. Materials are procedural cues and semantic priors, with optional observed color applied only to predefined observable roles; they are not textures or evidence of visual truth. The artifact package includes a visual-material audit for the six paper SPPA renders, checking that OBJ materials are covered by manifests, rendered views are non-empty, and the renderer uses material manifests with depth-sorted face drawing. The actor-based Unreal implementation creates many primitive components, so draw calls and component counts can become limiting before triangles do. The new opt-in HISM route reduces this pressure substantially in the packaged benchmark and is now selectable as a live backend. Its basic `descriptor_id`/action contract and group-level material/color behavior have regression tests. Its query-collision policy is also regression-tested for confirmed versus tentative obstacles. A per-instance custom-data path now avoids extra HISM groups for heterogeneous color/confidence attributes, and a cooked material now reads the per-instance color/alpha channels in the opt-in path. The optimized HISM path now has packaged desktop evidence that draw-group pressure can stay low at 100–500 synthetic objects. Dense all-object updates remain too expensive: subsequent no-CSV-profiler reruns put the 100-object shape stream above the 30 FPS P95 target. The new partial-update contract improves the claim only

under a changed-track scheduler: a synthetic 10% pose/shape schedule stayed below 33.3 ms P95 through 500 objects, while dense pose updates still failed at 250 and 500 objects. At 500 actors, the HTTP replay remains not real-time. These are not cosmetic limitations; they define the next research work.

SPPA also introduces a human-factors risk. A clean proxy can look more certain than the detector, georeferencing, yaw estimate, or scale estimate. The descriptor therefore exposes uncertainty fields, but the visual encoding must be tested. Until then, an “uncertainty-marked generic fallback” means a specified display behavior, not a proven operational guarantee.

9 Threats to Validity

The main threats are methodological rather than cosmetic. First, mask and part-adaptation diagnostics use synthetic geometry, so they do not establish real UAV silhouette recovery. Second, the neural-generator comparison uses proxy RGBA crops or speed-oriented text prompts, so it evaluates substitutability under controlled inputs rather than general generator quality. Third, Unreal evidence comes from internal packaged and replay tests without a headset, phase-aligned GPU-profiler trace, dense mission scene, flight-derived changed-track rates, or live network loop. Fourth, the finite ontology can resolve labels into reviewed families, but it cannot produce correct geometry for arbitrary words. Fifth, SPPA reduces triangle counts in some cases but can increase component and draw-call pressure; the HISM diagnostic reduces that pressure in packaged replay and partial-update benchmarks, but does not yet establish dense VR readiness or live-mission scalability. Sixth, the YOLOE open-vocabulary probe is a preliminary detector source, not a proof of universal recognition. Its role in this paper is to supply imperfect compact semantic evidence that SPPA can normalize; target-drone latency, false-positive sweeps, and domain-specific fine-tuning remain required before flight use.

10 Claim Boundaries

The supported claim is narrow. Sparse semantic labels can be compiled into bounded proxy descriptors; those descriptors can be consumed by an optional Unreal backend that coexists with curated assets; and pose plus matching-topology shape updates can be applied on the tested paths without regenerating the actor topology. The current evidence also supports a semantic normalization layer that turns approximate detector labels into reviewed proxy families or explicit conservative fallback, plus internal role-preservation diagnostics for synthetic vehicle dimensions and synthetic rectangle-mask metadata.

The manuscript does not claim end-to-end UAV perception validation, universal object recognition, dense VR readiness, operator readability, network advantage, photorealistic asset generation, real-image silhouette recovery, or operational risk reduction. Those are future validation targets, not hidden conclusions.

11 Conclusion

SPPA is best framed as a deterministic semantic rendering contract for UAV digital twins, not as a general 3D generator and not as the flight system. Its current evidence supports a narrow claim: object-level telemetry can be compiled into low-polygon, part-based proxy descriptors; those descriptors can be ingested by an optional Unreal backend while preserving the existing asset backend; and pose plus matching-topology shape updates can be applied without regenerating components in the tested paths. The ambitious version of the system is therefore not “perfect 3D from anything”; it is YOLOE-style approximate detection, SPPA semantic normalization, and reviewed/fallback proxy families that stay lightweight, controllable, and explicit about uncertainty.

The paper is not yet evidence for a deployed UAV/VR system. The strongest next step is not more textual justification, but a live or recorded semantic-telemetry replay that feeds the same detections into both curated assets and SPPA, followed by packaged Unreal/VR profiling and a controlled operator study. Until those artifacts exist, SPPA should be judged as a specification plus preliminary runtime evidence for a semantic proxy representation layer.

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